

Experiment No - 1

Object - Determination of efficiency of a single-phase transformer by a load test.

Apparatus Required -

S.No	Name of Apparatus	Specifications
1.	Digital Voltmeter	0-300v
2.	Digital Ammeter	0-5 A
3.	Digital Wattmeter	5 Amp, 300v
4.	Auto Variable Transformer	0-240 V
5.	Suitable Electrical Load	
6.	Connecting wires	As required.

Brief Theory -

Efficiency - The efficiency of a transformer is given by the expression -

$$\% \eta = \frac{\text{Output power}}{\text{Input power}} \times 100$$

$$= \frac{V_2 I_2}{V_1 I_1} \times 100$$

Where, W_1 = Power input to the transformer
 V_2 = Terminal Voltage across secondary at a particular load.

I_2 = Load Current.

Procedure —

1. Connect the circuit as per the circuit diagram shown in Fig 1.
2. Ensure that there is no load on the secondary windings of the transformer.
3. Switch on the A.C supply and record the no of load voltage across the secondary winding.
4. Adjust approximately 10 percent of full load current in the secondary by switching on certain lamps in the lamp bank load. Record the readings of all the meters.
5. Repeat step 4 for various load currents, till the full load value.

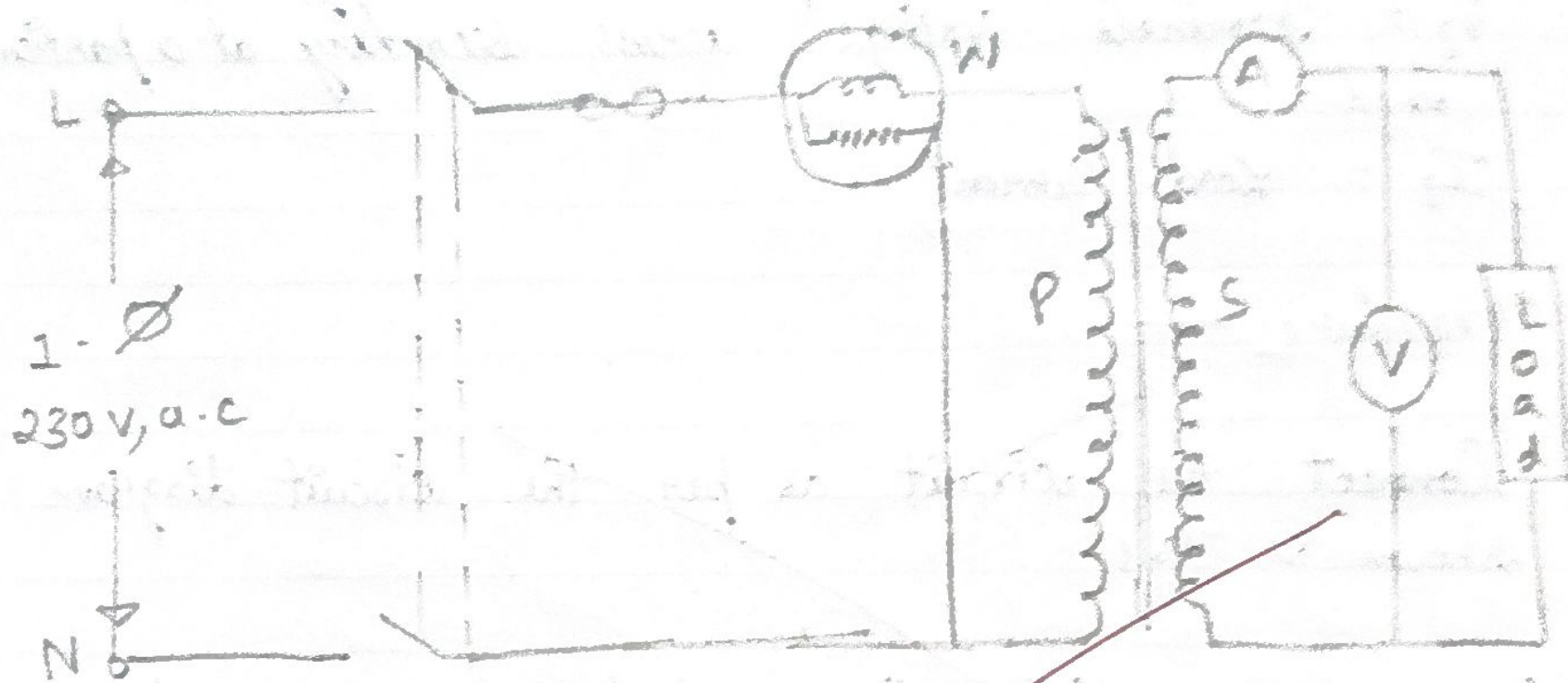


Fig-1

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6. Reduce the load on the transformer by switching off the bulbs in the lamp bank load.

7. Switch off the AC supply.

Observation Table -

S.No.	W_1	V_2	I_2	$V_2 I_2$ (W_2)	$\% \eta = \frac{V_2 I_2}{W_1} \times 100$
1.	229.3	223.2	0.89	198.64	86.6%
2.	427.3	219.2	1.77	387.99	90.8%
3.	522.9	217.2	2.20	477.84	91.3%
4.	707.4	214.0	3.04	650.56	90.19 91.96%

Average - $\frac{86.6 + 90.8 + 91.3 + 91.96}{4}$

= 90.16%

Calculation -

$$\% \eta = \frac{W_2}{W_1} \times 100$$

Teacher's Signature _____

Result -

Efficiency of the transformer is 90.16%.

Precaution -

1. All connections should be tight. -
2. All apparatus should be of suitable range and rating
3. Never touch the live terminals.
4. Under no-load condition, switch off the input supply.

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